

Day 2 – Power Supply Systems

Topic: SMPS, UPS and PC Power Connectors

Task 1: SMPS and UPS Devices

Devices that use SMPS

- 1. Desktop Computer:** It uses SMPS to convert AC power into different DC voltages needed by the motherboard, CPU, storage and other components.
- 2. LED Monitor:** The monitor uses an SMPS to convert AC supply into the lower DC voltage needed by its electronic circuits.
- 3. Wi-Fi Router:** A router uses a small power supply to convert AC electricity from the wall into DC power for its internal components.

Devices that use UPS

- 1. Desktop Computer:** A UPS provides backup power during a power cut, allowing the computer to be saved and shut down safely.
- 2. Wi-Fi Router:** A UPS can keep the router working during a power cut, helping maintain the internet connection for some time.

Task 2: Main Internal Components of a Typical SMPS

AC Input	→	EMI Filter	→	Rectifier	→	DC Filter
DC Filter	→	Switching Circuit	→	Transformer	→	Secondary Rectifier
Secondary Rectifier	→	DC Output Filter				

AC Input: Takes electricity from the wall socket.

EMI Filter: Reduces unwanted electrical noise.

Rectifier: Converts AC into DC.

DC Filter: Smooths the rectified DC voltage.

Switching Circuit: Switches the DC at high frequency.

Transformer: Changes the voltage to the required level and provides isolation.

Secondary Rectifier: Converts the transformed signal into DC.

DC Output Filter: Smooths the output before it is supplied to PC components.

Feedback Circuit: Monitors the output voltage and helps keep it stable.

Task 3: Power Connectors

Connector	Used for	Safety tip
ATX 24-pin	Main power for the motherboard.	Align it correctly and do not force the connector.
SATA Power	Power for SATA HDDs, SSDs and some SATA devices.	Turn off and unplug the PC before connecting or disconnecting it.
Molex 4-pin	Older HDDs, optical drives, fans and some accessories.	Use the correct connector while removing it; do not pull the cable.
PCIe 6/8-pin	Extra power for graphics cards (GPUs).	Use the connector required by the GPU and make sure it is fully inserted.

Note: Connector photos can be added here or kept in the GitHub images folder as requested in the assignment.

Task 4: Connecting SMPS in a Gaming PC

1. Switch off the power supply and unplug the PC from the wall socket.
2. Install the SMPS in the PC case and fix it properly with the screws.
3. Connect the **24-pin ATX cable** from the SMPS to the motherboard.
4. Connect the **4+4 pin CPU/EPS cable** to the CPU power connector near the processor.
5. Install the graphics card and connect the required **PCIe 6-pin or 8-pin cable** from the SMPS to the GPU.
6. For a SATA SSD or HDD, connect a **SATA power cable** from the SMPS to the storage drive.
7. Connect other required power cables for compatible fans or accessories.
8. Check all connections again and make sure no cable is loose.
9. Connect the AC power cable to the SMPS and switch on the PC.

Main Cables Used

PC Component	SMPS Cable
Motherboard	24-pin ATX
CPU	4+4 pin EPS/CPU
Graphics Card	PCIe 6/8-pin
SATA SSD/HDD	SATA Power
Older accessories	Molex, if required